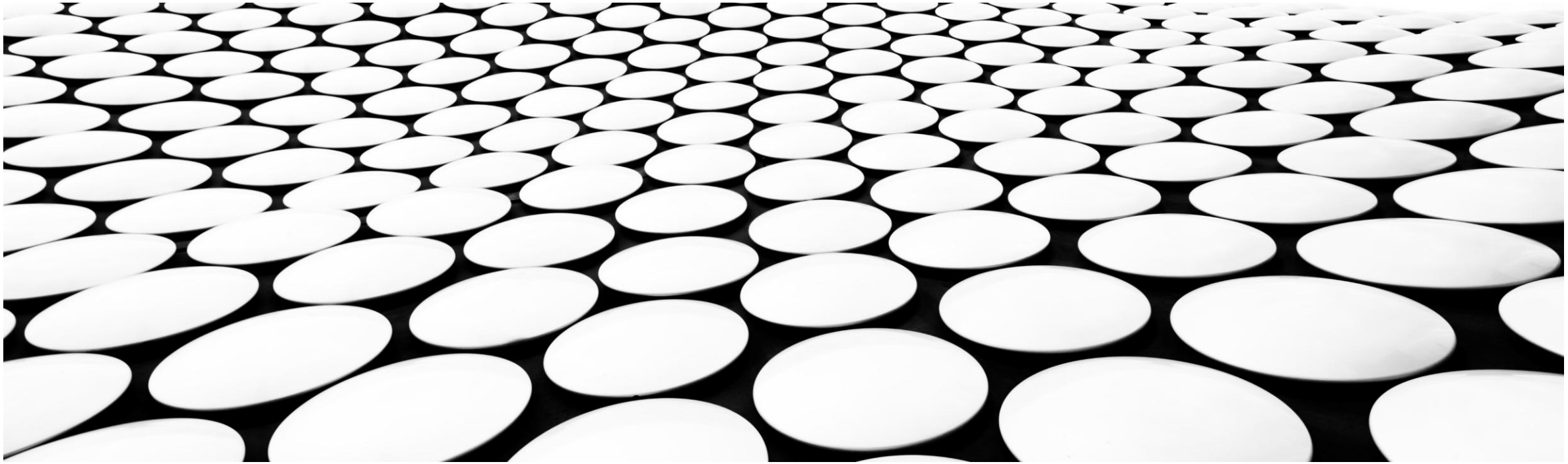
A Theoretical Study of Optical Limiter Designs Using Asymmetric Resonant Cavities with Sb_2Se_3 Layers — *Transfer-Matrix Method Approach* —

ADVANCED PHOTONIC LABORATORY



INTRODUCTION & MOTIVATION

Antimony triserenide

Key Advantages of Sb_2Se_3 for Optical Limiting

- Sb_2Se_3 offers exceptional properties for optical limiter applications, including ultra-low optical losses ($k < 10^{-5}$) in the telecommunication C-band, large refractive index contrast ($\Delta n = 0.77$) between phases, and excellent switching endurance (up to 10^7 cycles demonstrated).

1520 - 1570 nm

- The material exhibits strong nonlinear optical response with third-order susceptibility values around 10^{-9} esu and nonlinear refractive indices of 10^{-6} cm^2/W.

STEP 1 – LITERATURE REVIEW & MODEL SETUP

Dataset: 7 A New Family of Ultralow Loss Reversible Phase Change Materials for Photonic Integrated Circuits Sb2S3 and Sb2Se3

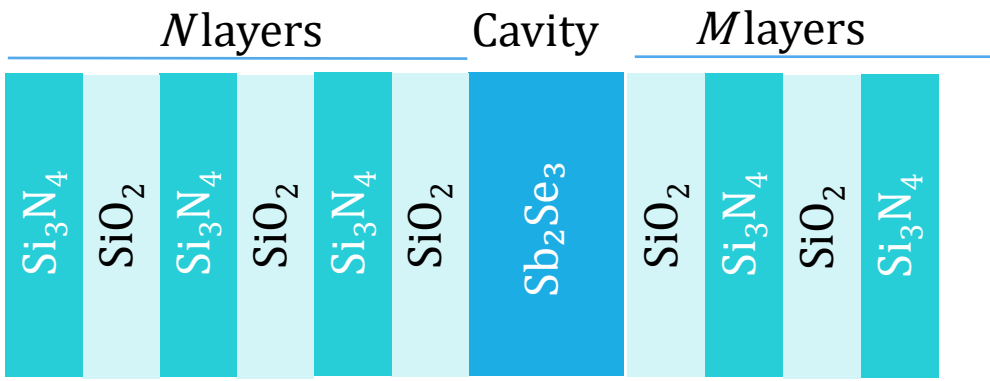
- Review of Sb_2Se_3 optical properties (n, k) for amorphous vs. crystalline states.

C-band of optical communication

- Selection of relevant wavelength range (e.g., 1520–1570 nm).
- Design of cavity geometry:
  - Sb_2Se_3 layer thickness
  - Dielectric spacers
  - DBR mirror specifications (e.g., SiO_2/Si_3N_4 DBRs)

In place of TiO2

- Boundary conditions: asymmetric Fabry-Perot cavity with unequal mirror reflectivity.



STEP 2 – TRANSFER-MATRIX MODEL CONSTRUCTION

- Fundamentals of TMM for multilayer optical structures
- Model implementation: input layer data (thickness, n, k)
- Encoding phase change: switching Sb_2Se_3 optical constants

STEP 3 – CAVITY SPECTRAL RESPONSE SIMULATION

check whether the current TMM calculation program can take angle input.

- Calculation of transmission/reflection/absorption spectra vs. wavelength and angle

The current AsymReca take resonance peak as input.

- Identification of resonance peaks, Q factors, and transmission contrast for phase transitions

Q factors = how effective the resonance is.

Optional work:

- Exploration of gradient thickness and multilayer setups for enhanced control

STEP 4 – NONLINEAR & LIMITING BEHAVIOR ANALYSIS

- Incorporation of intensity-dependent refractive index in Sb_2Se_3: Nonlinear refractive index for Kerr-type effects: $n(I) = n_0 + n_2 I$

  n = n0 + n2*(E(t))^2
  n = n0 + 2n2*|E(w)|^2

- Simulation of nonlinear transmission suppression at high incident intensities

- Demonstration of threshold switching and limiter action

STEP 5 – OPTIMIZATION & PARAMETRIC STUDIES

- Systematic variation of:
  - Sb_2Se_3 thickness
  - DBR layer number Incident -> M -> Sb2Se3 -> N -> Substrate
  - Mirror asymmetry degree
  - Cavity length
  - Refractive index values
- Performance optimization: targeting contrast, loss, and switching speed

Optional work:

STEP 6 – ENDURANCE & MULTILEVEL EFFECTS

How?

- Modeling of repeated switching cycles and stability analysis
- Simulation of multi-state phase transitions by interpolation of n and k
- Prediction of long-term durability and multi-level optical limiting behavior

STEP 7 – VALIDATION & BENCHMARKING

- Comparison with experimental results and literature simulation data (Sb_2Se_3 and other PCMs)
- Analysis of device stack compatibility for CMOS and photonic integration
- Potential deployment in practical on-chip limiter and modulator systems

[2] NICHOLAS A. NOBILE, et al., "Nonvolatile Tuning of Bragg Structures Using Transparent Phase-Change Materials"

SUMMARY & OUTLOOK

- Systematic TMM approach enables versatile design and simulation of Sb_2Se_3-based limiters
- Design recommendations for high-performance, robust, and low-loss photonic devices
- Future directions: experimental validation, hybrid control schemes, integration into photonic platforms